exact power of 2 ?

$$rax \overset{?}{=\!=} 2^{k \in N} \begin{cases} \xrightarrow{Y} ZF = 1 \\ \xrightarrow{N} ZF = 0 \end{cases}$$

```
mov     rdx,rax
dec     rdx
xor     rax,rdx
shr     rax,1
cmp     rax,rdx
```

$rdx := rax_0$

$rdx = rax_0 - 1$

$rax = rax_0 \oplus rdx$

$\hookrightarrow rax_0 \oplus (rax_0 - 1)$

$rax_* = (rax_0 \oplus (rax_0 - 1)) >> 1$

cmp

$rdx_* = rax_0 - 1$

empirical examples

$\overset{?}{=} \; rdx_* = rax_0 - 1 \overset{ZF=1}{=\!=\!=} (rax_0 \oplus (rax_0 - 1)) >> 1 = rax_*$

$\Updownarrow$

$rax_0 = 2^k \; ; \; k \in N$

ELSE $ZF = 0$

$f(5,7,13) \longrightarrow 0$

$f(1,2,4,8) \longrightarrow x - 1$

$f(12,14) \longrightarrow > 0$

mov	rdx, rax	rdx := rax
dec	rdx	rdx = rax − 1
xor	rax, rdx	
shr	rax, 1	≈ rax / 2
cmp	rax, rdx	rax $\overset{?}{=}$ rdx

∴ bool: (non-zero power of 2 ?) → Y/N

✓ 0001
0000
0001

✓ 0010 (2)
xor 0001
0011

✗ 0000
1111
1111

(12) 1100
xor 1011
✗ 0111 (7)
3

(13) 1101
1100
✗ 0001 (1)
0

(5) 0101
0100
0001 (1)
0

1110 (14)
1101
✗ 0011 (3)
1

0111 (7)
0110
✗ 0001
0

✓ 0100 (4)
0011
0111
3

(8) 1000
✓ 0111 (7)
1111 (15)
7

5
7
13 → 0

14 → 1

12, 4 → 3

8 → 7

5, 4, 2, 1